Final Exam

● Graded

Student

Yixuan Zhang

Total Points

44.5 / 45 pts

Question 1

Overall Grades **44.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 0.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Yixuan Zhang	yz1004	Yixuan Zhang

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Yixuan Zhang

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	5	5	5	44.5

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

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The system can be write as matrix $\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$

$R_3 \leftrightarrow R_1$ $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right]$ $R_3 \leftarrow R_3 - 2R_1$ $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$

$R_3 \leftarrow 2R_3 - 3R_2$ $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & 1 & -5 \end{array}\right]$ $R_1 \leftarrow R_1 + 2R_3$ $R_2 \leftarrow R_2 + 3R_3$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & -2 & 0 & -14 \\ 0 & 0 & 1 & -5 \end{array}\right]$

$R_2 \leftarrow -\frac{1}{2}R_2$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$

So. $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

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Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) ~~Col(A)~~ $A = [a_1 \; a_2 \; a_3]$

~~$a_1 = \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}$ $a_2 = \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}$ $a_3 = \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix}$~~

$A \xrightarrow{R_2 \leftarrow R_2 + 5R_4} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 15 & 15 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_1} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 15 & 15 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$

$\sim \begin{bmatrix} 2 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ pivot columns: $\begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}$

So Col(A) has basis $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

(2) Non-zero rows: $[2 \; 4 \; 2]$, $[0 \; 1 \; 1]$ -0.5

So Row(A) has basis $\{ [2 \; 4 \; 2] \; [0 \; 1 \; 1] \}$

(3) for $Ax = 0$ equals to $\left[\begin{array}{ccc|c} 2 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

let $x_3 = t$

$x_2 = -t$

$x_1 = t$

$x = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ Null(A) has basis $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

We can know A is a 2×3 matrix.

let $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

We know $T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}$, $T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}$, $T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}$

So, $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\cdot\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$, $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\cdot\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$, $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$

$\begin{cases} b-c=1 \\ e-f=1 \end{cases}$ and $\begin{cases} a-c=0 \\ d-f=2 \end{cases}$ and $\begin{cases} -c=2 \\ -f=2 \end{cases}$

So $\begin{cases} a=-2, b=-1, c=-2 \\ d=0, e=-1, f=-2 \end{cases}$

$A = \begin{bmatrix} -2 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$

Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

$\det(A-\lambda I) = \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda)(1-\lambda) - 0 + 0$

$= (1-\lambda)^2(2-\lambda) = 0$

$\lambda_1 = 1$, $AM_1 = 2$

$\lambda_2 = 2$, $AM_2 = 1$

for $\lambda_1 = 1$, $A\vec{v}_1 = \lambda_1\vec{v}_1$ can get $\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{array}\right]$

$R_1 \leftrightarrow R_3 \sim \left[\begin{array}{ccc|c} -2 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{array}\right]$ $R_3 \leftarrow R_1 + R_3$ $\left[\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{array}\right]$ $R_3 \leftarrow R_3 - 2R_2$ $\sim$ $R_1 \leftarrow -\frac{1}{2}R_1$ $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

Let $v_3 = t$, $\vec{v}_3 = t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $GM_1 = 1 < AM_1$

So the Jordan block $= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $J = \begin{bmatrix} J & 0 \\ 0 & 2 \end{bmatrix}$

for $\lambda_2 = 2$, similarly, $\left[\begin{array}{ccc|c} -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & -2 & -1 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} -1 & 2 & 0 & 0 \\ 0 & -6 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

Let $v_3 = t$, $\vec{v} = t\begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}$

Pick $\vec{v}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}$

For Jordan chain: $(A-\lambda_1 I)\vec{v}_2 = \vec{v}_1$, $(A-\lambda_1 I)^2\vec{v}_2 = (A-\lambda_1 I)\vec{v}_1 = 0$

$A - \lambda_1 I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$, $(A-\lambda_1 I)^2 = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & -6 & 0 \end{bmatrix}$, and $-2v_1 - 2v_2 = 0$

for $\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -6 & 0 & 0 \end{array}\right]$ pick $\vec{v}_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ -1 \end{bmatrix}$

So Jordan Canonical form

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$f(x,y,z) = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}.$$

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix} = (3-\lambda)\left((4-\lambda)(3-\lambda)-1\right) - 0 + \left(\lambda - 3\right)$$

$$= (3-\lambda)\left((4-\lambda)(3-\lambda) - 1\right)$$

$$= (3-\lambda)\left(\lambda^2 - 7\lambda + 10\right) = (3-\lambda)(\lambda-5)(\lambda-2)$$

$$= 0$$

So $\lambda_1 = 3$, $\lambda_2 = 5$, $\lambda_3 = 2$, AM = 1 for each

for $\lambda_1 = 3$, $(A-3I)\vec{v}_1 = 0$

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \text{ pick } \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

for $\lambda_2 = 5$, $(A-5I)\vec{v}_2 = 0$

$$\left[\begin{array}{ccc|c} -2 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 1 & -1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} -2 & 0 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \text{ pick } \vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

for $\lambda_3 = 2$, $(A-2I)\vec{v}_3 = 0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \text{ pick } \vec{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

So, $D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $P = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 0 & -2 & 1 \end{bmatrix}$

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Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

It can be write as $\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \leftarrow 2R_3 - R_2} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & 3 & 3 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow \frac{1}{3}R_3} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

So $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ $v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$, obviously v_1, v_2 are linearly independant

So $\{v_1, v_2\}$ is a basis for V.

Orthogonalize:

$$u_2 = v_2 - \frac{v_1^T v_2}{v_1^T v_1}\cdot v_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{0\times3+1\times0+1\times2+0\times(-1)}{0\times0+1\times1+1\times1+0\times0}\cdot\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

Check, $v_1^T \cdot u_2 = 0$

Normalize:

$\|v_1\|^2 = 2$, $\|u_2\|^2 = 9+1+1+1 = 12$

$\|v_1\| = \sqrt{2}$, $\|u_2\| = 2\sqrt{3}$

$$\frac{v_1}{\|v_1\|} = \begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix} \qquad \frac{u_2}{\|u_2\|} = \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}$$

So $\left\{\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\end{bmatrix}\right\}$ is an orthonormal basis for V.

Problem 8. [8] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

For $n \times n$ matrix A, $A^T = -A$, $A = -A^T$

$\det(A) = \det(-A^T) = (-1)^n \det(A^T)$

$= (-1)^n \det(A)$

for n is odd: $(-1)^n = -1$

$\det(A) = -\det(A)$

$2\det(A) = 0$

So, $\det(A) = 0$

there must: $\text{diag}(A) = [0, \cdots, 0]$

$[A_{ij}]^T = -A_{ij}$

$A_{ji} = -A_{ij}$

$$A = \begin{bmatrix} 0 & A_{12} & \cdots & A_{1n} \\ A_{21} & 0 & \ddots & \vdots \\ \vdots & \ddots & 0 & A_{n-1\,n} \\ A_{n1} & \cdots & A_{n\,n-1} & 0 \end{bmatrix}$$

$$\det(A) = 0 + \sum_{i=2}^{n} (-1)^{i+1} A_{1i} (M_{1i})$$

There are $(n-1)$ elements in Summation Σ.

for each non-odd i:

$i \to (-1)^{i+1} A_{1i} (M_{1i})$

$i+1 \to (-1)^{i+2} A_{1\,i+1} (M_{1\,i+1})$

$A_{1i}(M_{1i}) = A_{1\,i+1}(M_{1\,i+1}) = \prod_{\substack{i=1,j=2, \\ i<j}}^{n} A_{ij}$

So for odd n, each ith elements must be compromised by $i+1$ elements

So, $\det(A) = 0$

Problem 9. [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

Square both sides: $\left(\|\vec{u}+\vec{v}\|\right)^2 = \left(\|\vec{u}-\vec{v}\|\right)^2$

$$(\vec{u}+\vec{v})^T \cdot (\vec{u}+\vec{v}) = (\vec{u}-\vec{v})^T \cdot (\vec{u}-\vec{v})$$

$$\vec{u}^T\vec{u} + \vec{v}^T\vec{u} + \vec{u}^T\vec{v} + \vec{v}^T\vec{v} = \vec{u}^T\vec{u} - \vec{v}^T\vec{u} - \vec{u}^T\vec{v} + \vec{v}^T\vec{v}$$

$$2(\vec{v}^T\vec{u} + \vec{u}^T\vec{v}) = 0$$

$$\vec{v}^T\vec{u} + \vec{u}^T\vec{v} = 0$$

and we know $\vec{v}^T\vec{u} = \vec{u}^T\vec{v}$ (dot product $\vec{v}\cdot\vec{u} = \vec{u}\cdot\vec{v}$)

So $\vec{v}^T\vec{u} = \vec{u}^T\vec{v} = 0$

$\vec{v}$ is perpendicular to $\vec{u}$